

Blockchain-based AMI framework for data security and privacy protection

Hongliang Tian^{a,b}, Yuzhi Jian^{b,*}, Xiaonan Ge^b

^a Key Laboratory of Modern Power System Simulation and Control & Renewable Energy Technology, Ministry of Education, Northeast Electric Power University, Jilin, China

^b School of Electrical Engineering, Northeast Electric Power University, Jilin, China



ARTICLE INFO

Article history:

Received 2 February 2022
Received in revised form 7 June 2022
Accepted 9 June 2022
Available online 15 June 2022

Keywords:

Blockchain
Advanced metering infrastructure
Power system security
Data security
Security framework
Privacy protection

ABSTRACT

As a key component of smart grid (SG), advanced metering infrastructure (AMI) realizes two-way communication between users and power companies. Considering that AMI involves a large number of important data, which are related to the measurement of users' electricity bills and the decision-making of power suppliers. Therefore, it is necessary to strengthen the protection of AMI data security and privacy to avoid economic losses caused by data leakage and tampering. Aiming at the problems of trust and single point of failure in the existing protection schemes, this paper proposes an AMI security framework based on blockchain. By combining the characteristics of blockchain such as decentralization, tamper proof, traceability, the architecture of AMI data management system is reshaped. The technologies of peer-to-peer (P2P) transmission, asymmetric encryption, consensus algorithm and smart contract of blockchain are used to solve the problems of secure storage, transmission and application of AMI data. Through theoretical analysis and experiments, the feasibility of introducing blockchain technology into AMI is verified. The results show that the proposed framework is secure and reliable, which provides a strong guarantee for AMI data security and privacy protection.

© 2022 Elsevier Ltd. All rights reserved.

0. Introduction

Smart grid plays a vital role in coping with global climate change, promoting energy conservation and emission reduction, and developing low-carbon economy. Through close integration with information and communication technology (ICT), smart grid realizes the automation of energy consumption billing, monitoring, planning and forecasting [1]. Compared with traditional power grid, the main advantage of smart grid is that it can provide nearly real-time two-way communication for power suppliers and end users, so as to better optimize energy production and consumption [2].

As an important part of SG, AMI inherits the two-way communication ability of smart grid and brings new opportunities for consumers and power suppliers [3]. Consumers can use real-time energy pricing to change their way of using electricity and reduce the cost of living. Power suppliers can realize the functions of remote meter reading, automatic power-off management, remote connection and disconnection through AMI [4].

AMI not only brings great benefits to users and power suppliers, but also provides criminals with new opportunities to

manipulate energy consumption data and control command. On the one hand, because most AMI components are installed outside, such as data concentrator, they often suffer from human damage and mechanical failure, so the single point of failure problem is difficult to avoid [5]. On the other hand, since the communication links used for data interaction in AMI are also exposed, network attacks from hackers will pose a great threat to AMI data security and privacy [6].

In response to the above problems, many schemes have been proposed to improve AMI data security and privacy protection [7–9]. Naseer et al. [7] attempted to provide guarantee for AMI secure communication by improving the security of keys used to encrypt transmitted data. However, this scheme has high communication overhead, so it is difficult to be applied to AMI system with limited component resources. Yu et al. [8] proposed a key management system based on information centric networking (ICN), which can meet various secure communication modes of AMI. However, this scheme relies too much on centralized components, so there is a serious single point of failure problem. Wang et al. [9] introduced blockchain technology, but in his scheme, blockchain only exists in the form of a central database, which fails to give full play to the distributed characteristics of blockchain. In short, the existing AMI data security and privacy protection schemes still have large room for

* Corresponding author.

E-mail addresses: hltian@foxmail.com (H. Tian), yzjian96@163.com (Y. Jian), gexiaonan1995@163.com (X. Ge).

improvement [10]. Therefore, we propose an AMI security framework based on blockchain, and try to combine key technologies [11] such as P2P network transmission, consensus algorithm and smart contract to make up for the defects and shortcomings of the existing AMI framework. The main contributions of this paper are as follows:

- (1) This paper comprehensively discusses the application of blockchain technology in AMI, and highlights the advantages brought by introducing it into AMI.
- (2) An AMI security framework based on consortium blockchain is proposed, which can not only provide security and privacy protection for AMI data, but also reduce the impact caused by single point of failure.
- (3) The specific role and operation mechanism of each layer of the proposed framework are discussed hierarchically, which provides a new perspective on how to better apply blockchain technology to AMI.

The rest of this paper is organized as follows. In Section 1, we review the related work. Section 2 introduces AMI and blockchain respectively, and discusses their compatibility. Section 3 proposes a theoretical framework and elaborates the operating mechanism of the framework. Sections 4 and 5 verify the security and feasibility of the proposed scheme respectively. Finally, Section 6 gives a summary of this work.

1. Related work

In order to ensure the security and privacy of AMI data, scholars at home and abroad continue to put forward new solutions, which mainly focus on key management, data aggregation and identity authentication.

Faroop et al. [12] proposes a key protocol based on identity authentication, which combines elliptic curve bilinear pairing algorithm to generate session key. AMI data is transmitted in ciphertext by session key to ensure the security and privacy of AMI data. Ruj et al. [13] proposed a distributed data aggregation and access control method, combined with homomorphic encryption algorithm to ensure that the data is transmitted and aggregated in the form of ciphertext, which greatly reduces the exposure of plaintext. Tong et al. [14] proposed an AMI data aggregation scheme based on data recovery. This scheme combines homomorphic encryption and chinese remainder theorem (CRT). Mohammadali et al. [15] believe that the availability of AMI data is related to performance, and confidentiality and integrity can be achieved by means of key encryption. Considering the high load of PKI certificate management, they proposed an identity based key management protocol.

Kumar et al. [16] emphasized the importance of network security in AMI communication and pointed out that key management can effectively ensure the confidentiality and integrity of AMI data. Considering that most of the existing key management schemes adopt centralized architecture and rely excessively on individual entities for key distribution and updating, they propose an efficient distributed multicast key management protocol. Zhang et al. [17] proposed a distributed security keyless strategy based on blockchain. In this strategy, intelligent devices send requests and receive responses through blockchain network for data transmission operations, and these operations are recorded in the form of immutable logs. Lu et al. [18] proposed a lightweight edge blockchain-assisted data aggregation strategy for smart grid data security and privacy protection. This strategy can not only effectively filter fake data from the user layer, but also improve the robustness of the system.

The scheme described in [12] needs to rely on a third-party trusted institution, so there is a serious trust problem and single-point fault problem in this scheme. Although multiple key distribution centers are set in the scheme described in [13], the single point fault problem has not been completely solved. The AMI communication framework described in [14] is a typical centralized architecture. Once an AMI component fails, the whole aggregation system will be paralyzed. The scheme described in [16–18] improves the security and robustness of the proposed scheme by combining blockchain technology, but lacks a comprehensive and systematic analysis of the AMI framework based on blockchain. In addition, some of their schemes simply use blockchain as a database or contract actuator, fail to make full use of system component resources and give full play to the decentralization characteristics of blockchain.

2. Demand analysis

In this section, we first introduce the basic composition of AMI. Secondly, after fully considering the threats and challenges faced by AMI data, we comprehensively discussed the advantages of introducing blockchain technology into AMI. Finally, through comparative analysis, we get the conclusion that consortium chain is more suitable for AMI than public chain and private chain.

2.1. Traditional AMI framework and its security problems

AMI is usually composed of smart meters (SM), data concentrators (DC), data management systems (MDMS) and communication networks among components.

- (1) SM: a data acquisition and receiving device, which sends the detailed energy consumption data of the user end to the data concentrator according to the established frequency.
- (2) DC: in order to realize the data forwarding between the IP network and the non-IP network, the data concentrator is not only used to collect the information data collected by the smart meters in the jurisdiction, but also to maintain the communication link between the smart meters and the power companies.
- (3) MDMS: it is used to store and analyze the data uploaded by the DC, issue control instructions according to the data analysis results, and maximize the benefits of power grid assets through the rational allocation of intelligent devices.

With the increase of the number of smart meters, the security problems related to AMI have increased significantly both internally and externally. On the one hand, the detailed information of customers' energy consumption is crucial because it can reveal their lifestyles. By long-distance transmission of data and retransmission or analysis of data stored in different places, it may also cause loopholes in data tampering, leaks or operations. On the other hand, the price signals and control commands received by consumer terminals are also potential areas of network and physical attacks, whose ultimate purpose is to destroy power infrastructure or implement power theft [19].

2.2. Compatibility analysis of blockchain technology and AMI

Because AMI involves a large number of data related to the measurement of users' electricity bills and the decision-making of power suppliers, it has high requirements for data security and privacy protection. The emergence of blockchain technology can well solve the problems AMI is currently facing. It ensures the security and privacy protection of data in the storage and

Table 1
Compatibility analysis of blockchain and AMI.

Feature	AMI	Blockchain+AMI
Security	AMI data involves the vital interests of users and power companies. The disclosure and tampering of data will cause significant economic losses.	The blockchain adopts asymmetric encryption to sign, encrypt and verify the transmitted data, so as to improve the security of AMI data in the process of transmission and application.
Privacy	Fine-grained energy consumption data can capture user behavior after detailed "consumer analysis", such as what type of appliances are used, and personal privacy will be exposed.	The blockchain adopts hash encryption algorithm to hide data source, data content and other information. In addition, through the setting of access control policy, data access is restricted to improve data privacy.
System structure	With the change of power grid structure, the integration of distributed energy and the emergence of producers and sellers, the traditional centralized architecture will face the problems of limited transaction processing performance, flexibility and single point failure.	The blockchain adopts P2P network transmission, distributed storage and consensus mechanism to ensure that the normal operation of AMI will not be interrupted due to some faults or malicious nodes. In addition, through the deployment and invocation of smart contracts, the transaction processing efficiency is significantly improved.

Table 2
Types and characteristics of blockchain.

	Public blockchain	Consortium blockchain	Private blockchain
Participant	Free entry for anyone	Consortium member	Within an individual or company
Consensus mechanism	POW/POS/DPOS	Distributed consistency algorithm	Distributed consistency algorithm
Bookkeeper	For all market participants	To be determined by the alliance members through consultation	User-defined
Incentive mechanism	Require	Optional	Optional
Degree of centralization	Completely decentralized	Polycentric	Polycentric or centralization
Prominent characteristic	Self-establishment of credit	Efficiency and cost optimization	Transparent and traceable
Transaction per second	3 to 20 transactions per second	1,000 to 10,000 transactions per second	1,000 to 200,000 transactions per second

transmission process through encryption authentication, authorization. Record all operations on data in the form of logs to ensure the authenticity and effectiveness of data. P2P transmission and consensus mechanism are adopted to ensure the unification of data on the chain and avoid single point of failure. Through the deployment of smart contracts, the ability of users or power service providers to interact with data has improved. In order to achieve a comprehensive link between blockchain and AMI, the compatibility between the two is described in detail as shown in Table 1.

2.3. Selection of blockchain type

According to the network coverage and the characteristics of participating nodes, blockchain can be divided into public chain, consortium chain and private chain [20]. A comparison of these three blockchain features is shown in Table 2.

Considering the characteristics of AMI data, such as large amount of data (smart meter generates millions of data information every day according to fixed acquisition frequency), so the selected blockchain type should have the characteristics of fast transaction processing speed and no need to consume "gas" similar to that in Ethereum. In order to ensure the data security and user privacy of AMI, the addition of components (such as data concentrators) in AMI should not be free and unlimited. Appropriate privilege control and access control management are of great significance to improve the overall security of AMI. Therefore, the selected blockchain type should have the characteristics of privilege control and access management. In summary, this paper will use consortium chain to reshape AMI data management system architecture.

3. Proposed framework

In this section, we have a hierarchical discussion of the proposed framework, which are data collection layer, network layer, consensus layer, contract layer and application layer. We then go into detail about the use of each layer and how it works.

The reason why this paper does not include the incentive layer mentioned by Li et al. [21] is that the participants in the consortium chain cooperate with each other for common benefits. Therefore, they have the motivation to contribute their computing power, storage space and network resource, and generally do not need to be motivated by additional coins. In the AMI security framework based on the consortium chain, each layer has its own core functions, as shown in Fig. 1.

3.1. Data collection layer

The data collection layer is the physical area and basic module of AMI. Its specific function is to regularly collect the energy consumption data on the user side. The collected metering data can be represented by Eq. (1):

$$m = \{ID_{SM}, T_M, P_C\} \quad (1)$$

Where ID_{SM} represents the device code, and note that it is unique. T_M represents metering time, P_C represents electricity consumption.

After the data acquisition, SM will send m to DC through the following steps:

Step 1: Calculate the hash value of the measurement data m : $H(m)$;

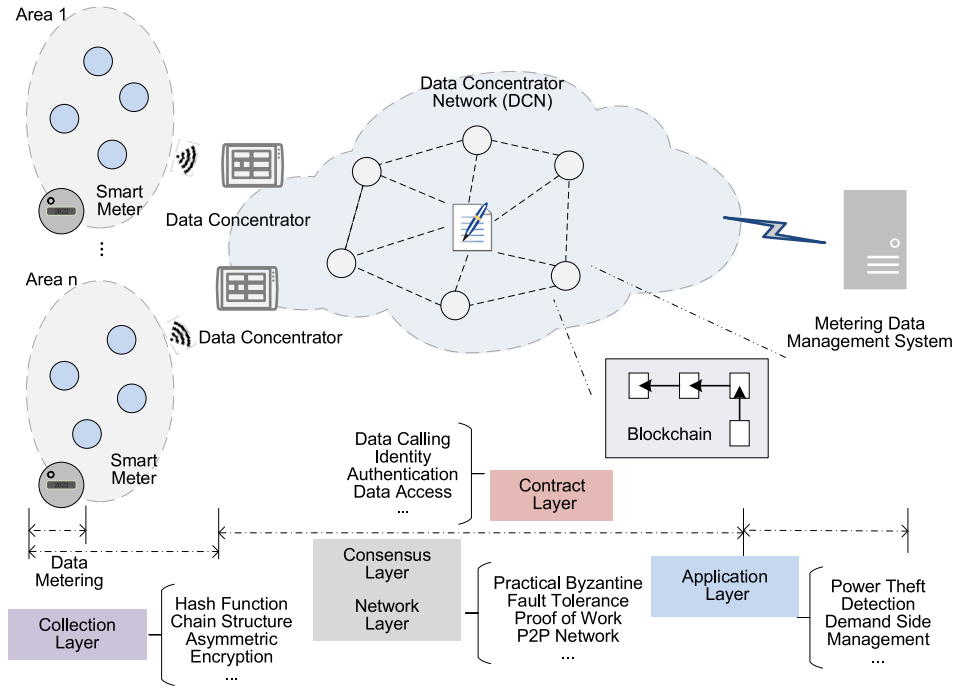


Fig. 1. AMI security framework based on consortium chain.

Step 2: SM encrypts m with the public key P_{DC} of DC to obtain the ciphertext $E(m)_{P_{DC}}$;

Step 3: SM sends message tuple $\{H(m), E(m)_{P_{DC}}\}$ to DC.

As shown in the above steps, the AMI data management platform combines asymmetric encryption technology to generate public-private key pairs, in which the private key is used to encrypt the transmitted data and the public key is used to decrypt the encrypted data.

After receiving the information from SM, DC will perform the following operation steps:

Step1: DC decrypts ciphertext $E(m)_{P_{DC}}$ with its private key to obtain plaintext information m .

Step2: Calculate the hash value $H'(m)$ of the plaintext information m using the same hash function used for smart meters. Compare $H'(m)$ and $H(m)$, if consistent, continue with the next steps.

Step3: Construct transaction T_x , as shown in Eq. (3):

$$Sign_{DC} = E(H'(m))_{S_{DC}} \quad (2)$$

$$T_x = \langle V_n, A, E(m)_{S_{DC}}, Sign_{DC} \rangle \quad (3)$$

Where V_n represents the number of the transaction, A is the area where smart meter is located, and $E(m)_{S_{DC}}$ is the encrypted measurement data using DC's private key.

Step4: During the specified time period, package multiple transactions into blocks and broadcast them to data concentrator network (DCN) for validation by other nodes. The composition and connection of blocks are shown in Fig. 2.

As shown in Fig. 2, the transactions verified by nodes in the whole network form a chain structure in chronological order in block. The block mainly includes block header and block body. The block header is mainly composed of block number, timestamp, hash value of parent block, merkel tree root, etc. The block body generally contains a list of transactions.

The above merkel tree root is generated by combining multiple hash operations and merkel tree construction, as shown in Fig. 2. It is usually used to ensure the integrity of data and the reliability of sources. DCN network nodes will check the legitimacy of transaction content when receiving new blocks. When anomaly

is detected, the method of tree traversal is used to quickly detect and locate abnormal transactions. The problematic transactions can be found by comparing at most $\log(N)$ times, where N represents the number of transactions of smart meters in a certain region.

3.2. Network layer

The network layer mainly includes the data transmission protocol, networking mode, data transmission and verification mechanism of the blockchain network.

- (1) Transmission protocol: the blockchain based AMI network layer adopts P2P transmission protocol. Under this transmission protocol, the nodes of the whole network have the same status and are connected with each other in a flat topology. Each network node has the functions of network routing, transaction verification and block propagation.
- (2) Networking mode: It can be seen from Fig. 1 that this networking mode is decentralized, and the blockchain network is constructed with the data concentrator as the node. Each node in the network can store, verify, and query the transmitted data without relying on other nodes.
- (3) Data transmission and verification mechanism: SHA-512 hash algorithm and digital signature technology are mainly used in data transmission and verification process.

a. SHA-512 secure hash algorithm is used to convert the input information into 128-bit hexadecimal string. For example, the unique identification of the node (such as equipment coding) outputs a 128-bit string through hash operation. Because hash function has the characteristics of fast forward calculation, difficult reverse calculation, sensitive input and strong anti-collision, it can ensure the uniqueness and concealment of information sender identity.

b. Considering that the encryption strength of 160-bit elliptic curve encryption (ECC) is the same as that of 1024-bit RSA encryption algorithm. Therefore, the ECC-based elliptic curve digital signature algorithm (ECDSA) is selected to improve the security

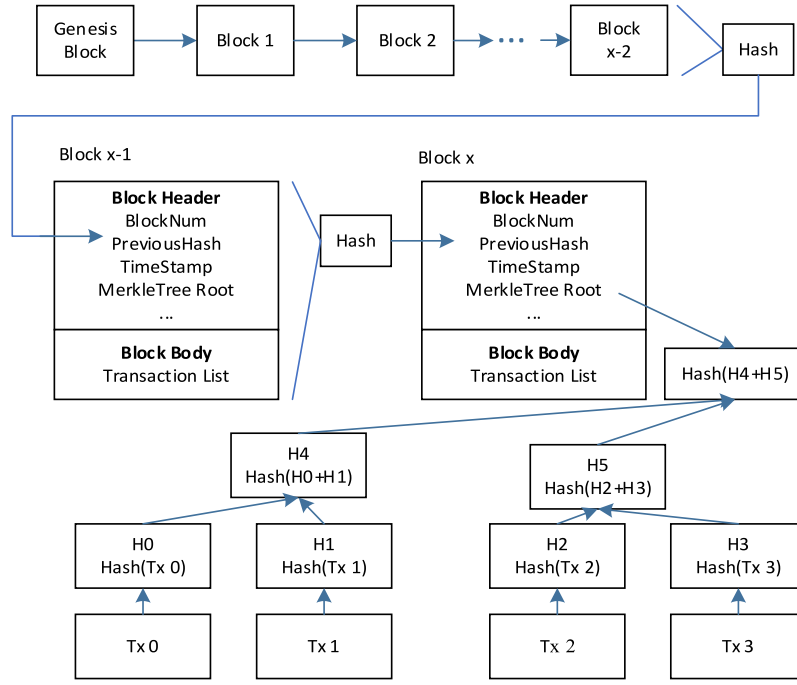


Fig. 2. Block structure and connection.

of data transmission in the case of occupying less storage space and network bandwidth.

Fig. 3 shows a use case of hash algorithm and digital signature, and the specific steps are as follows:

Step1: The sender calculates the digital summary of the original data through the hash algorithm, and uses its private key to encrypt the digital summary. The encrypted data are digital signatures.

Step2: The digital signature is sent together with the original data to the party that verifies the signature.

Step3: The verifier decrypts the digital signature with the sender's public key to obtain the original digest value.

Step4: The verifier calculates the summary value of the received original data by using the same hash algorithm as the sender. Compare the decrypted digest value with the recalculated digest value. If they are the same, the signature verification passes.

The sender's public key can decrypt the digital signature to ensure that the original data does come from the sender. The summary value obtained after decryption is the same as that obtained by recalculation of the original data, which ensures that the original data are not tampered with during transmission. After completing the verification of message integrity and sender identity legitimacy, the transaction will trigger the subsequent processing flow.

In DCN, the data concentrator is mainly responsible for aggregating the measured data from smart meters, and packaging the aggregated data into blocks in the form of transactions. The newly generated block is transmitted through P2P network. After receiving the block, other nodes will simulate the transaction locally, and check the validity and authenticity of the message according to the time stamp, Merkel tree root and other information. If the verification is passed, the block will be linked to the longest blockchain, and vice versa.

3.3. Consensus layer

The consensus layer includes various consensus algorithms. When the voting rights are highly dispersed, the block content

can be agreed through the consensus algorithm. Therefore, consensus algorithm is considered to be an indispensable part of blockchain technology. The application of consensus algorithm fundamentally solves the trust problem between nodes.

According to the balance between availability and consistency of consensus algorithm, it can be divided into two categories: absolute consistency algorithm and probabilistic consistency algorithm. Probabilistic consistency algorithms such as proof of work (POW) make the inconsistent data of each node disappear gradually with the passage of time. Absolute consistency algorithms such as byzantine fault tolerance (BFT) make the data between nodes highly consistent.

Considering fault tolerance and energy consumption, the practical byzantine fault tolerance (PBFT) algorithm is more suitable for AMI components with limited resources. It not only consumes fewer computing resources and time to promote the whole network nodes to reach a consensus, but also maintains high fault tolerance. As long as the number of malicious or faulty nodes does not exceed $1/3$ of the total nodes, the AMI data management system can operate normally. The following is a description of PBFT consensus algorithm:

The application premise of PBFT consensus algorithm is to use cryptography to ensure that the message transmission between nodes is reliable enough. PBFT is a state machine replication algorithm that replicates data copies on different data aggregator nodes. Assuming that the number of faulty network nodes is f , in order to ensure the normal operation of the system, $2f + 1$ normal nodes are required, that is, the total number of system nodes is $3f + 1$ [22].

As shown in Fig. 4, C is the client, M is the master node, N_1 , N_2 is the slave node, and N_3 is the malicious node. The consensus process is as follows:

- (1) Request: Select a node M from the network node, which is responsible for the generation of new blocks.
- (2) Preparing: M node receives and packages the requests sent from client C , and then broadcasts them to the whole network, namely, N_1 , N_2 , N_3 nodes.

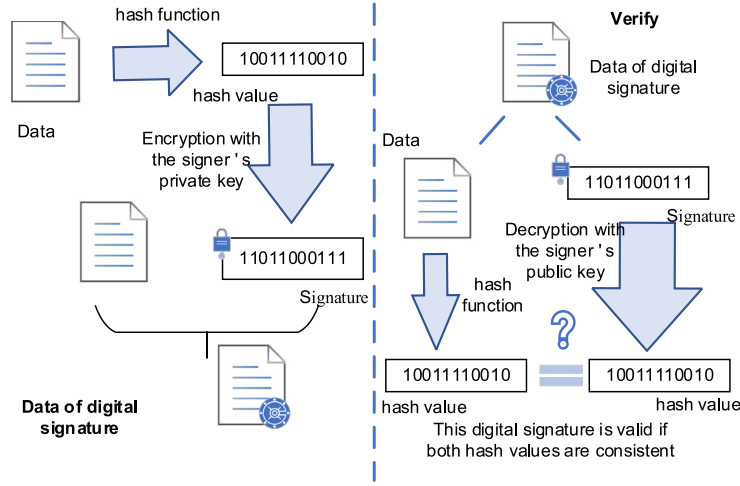


Fig. 3. Digital signature and verification.

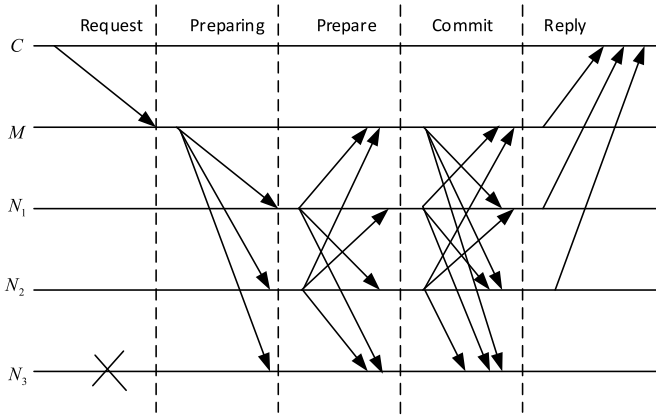


Fig. 4. PBFT consensus process.

- (3) Prepare: Each node simulates execution after receiving the transaction. Based on the execution results, the summary value of the new block is calculated and broadcast to the whole network, that is, $N_1 \rightarrow N_2, N_3, M, N_2 \rightarrow N_1, N_3, M$.
- (4) Commit: if a node receives $2f$ summaries from other nodes equal to its own, broadcast a commit message to the entire network.
- (5) Reply: when a node receives $2f + 1$ commit message, it means that the data information is confirmed.

3.4. Contract layer

Smart contract is an important part of the contract layer and the basis for flexible programming and data operation of the blockchain system. In the application of AMI data, routine operations such as reading and writing of data can be realized by calling intelligent contracts, or automatic upload of AMI data information and privilege management of data visitors can be realized by writing complex intelligent contracts.

Smart contract is the essence of contract layer, which has the characteristics of independence, reliability and high efficiency. At the initial stage of building the blockchain system, technicians need to formulate the content of the smart contract and the trigger conditions for contract execution, and embed it into the AMI data management system in the form of script code. When the system is running, once the smart contract meets the trigger

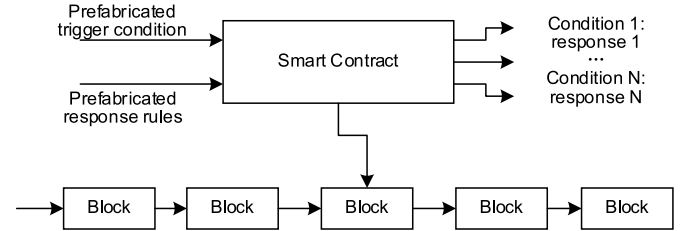


Fig. 5. Operation logic of smart contract in blockchain.

conditions, it will be executed automatically. Moreover, smart contracts will not be affected by external factors in the process of execution, so as to improve the efficiency of data interaction on the premise of fairness. Fig. 5 shows the operation logic of smart contract in blockchain.

3.5. Application layer

The application layer provides users with diversified services by flexibly calling the contract layer interface. By integrating and analyzing the power consumption data on the user side, the power supplier can determine the power consumption mode (such as high and low peak periods) and electricity consumption trend on the user side. Then, they take economic and technical measures to encourage users to use energy-saving technologies and change the way of energy consumption. In addition, power theft detection can be realized based on the data analysis results. When the power consumption on the user side is abnormally high, an alarm can be issued to reduce the loss of power enterprises.

Take power demand side management for example. MDMS reads user energy consumption data through access interfaces provided by blockchain, as shown in algorithm 1. After obtaining the area user data, MDMS analyzes the power data and formulates the time-sharing price to realize the purpose of load peak shaving and valley filling, thus producing obvious economic and social benefits.

4. Safety analysis

In this section, we will comprehensively demonstrate the security of the proposed scheme from three aspects, namely, transaction generation stage, confirmation stage and access stage. The results show that the proposed scheme can meet the safety requirements of AMI.

Algorithm 1: Read user energy consumption information

1. The MDMS administrator calls the query function and takes A as the input parameter;
2. Traverse the DCN database and query whether there are relevant transactions;
3. If A is retrieved, all relevant information tuples $T_x = \langle V_n, A, E(m)_{s_{DC}}, \text{Sign}_{DC} \rangle$ in the area are returned. If no relevant information is retrieved, an error is returned.

4.1. Security in transaction generation stage

Each operation request for AMI data can be seen as a transaction generation. In the transaction generation stage, timestamp information is one of the important parameters to deal with replay attacks. The application of ECDSA makes the attacker unable to obtain effective original data because of the lack of private key. In addition, the single hash function SHA-512 has the characteristics of input sensitivity (even if the input information changes slightly, the reconstructed hash value will be different from the original hash value), strong anti-collision (different input is difficult to obtain the same output) and so on, which improves the robustness of the system.

4.2. Security in the transaction confirmation phase

Kraft D. [23] proves that the attack process of hackers on power data platform meets the characteristics of Poisson distribution. Whether AMI data can be determined ultimately depends largely on the length of blockchain. Assuming that the probability of trust node producing the next transaction in AMI data system is p , the probability of malicious node producing the next transaction is q , and satisfies $p + q = 1$. Then the probability that the attacker can eventually eliminate the gap of n blocks is P_n .

$$P_n = \begin{cases} \left(\frac{q}{p}\right)^n, & p > q \\ 1, & p \leq q \end{cases} \quad (4)$$

The blockchain extension length of AMI data attackers satisfies Poisson distribution, and its expected value is λ .

$$\lambda = n \frac{q}{p} \quad (5)$$

Assuming that the existing block chain length is n , the attacker imitates the normal transaction and initiates forgery attack on AMI data, then the probability that the forgery transaction node can successfully tamper with the transaction is expressed as P_ζ :

$$P_\zeta = \sum_{k=0}^{\infty} \frac{\lambda^k e^{-\lambda}}{k!} \cdot \begin{cases} \left(\frac{q}{p}\right)^{n-k}, & k \leq n \\ 1, & k > n \end{cases} \quad (6)$$

In order to accurately quantify the potential risk of blockchain system, this paper calculates the probability P_ζ of forgery transaction node successfully tampering with AMI data by Matlab software. The Origin software is used to express the success rate of power grid data tampering. The test results are shown in Fig. 6. It can be seen from the figure that with the improvement of the attacker's ability (with high block forgery ability), the security risk of the system increases sharply. When the number of blocks is 2, $q = 0.3$, the transaction failure risk is close to half. In order to deal with the attacker's interference, the increase in the number of blocks can effectively reduce the success rate of the attacker tampering with the transaction [24]. When the attack node obtains the accounting probability $q = 0.5$, the power grid data can be tampered with no matter how many blocks exist in the system. In other words, if you want to illegally tamper with AMI data, you must have more than half of the computing power

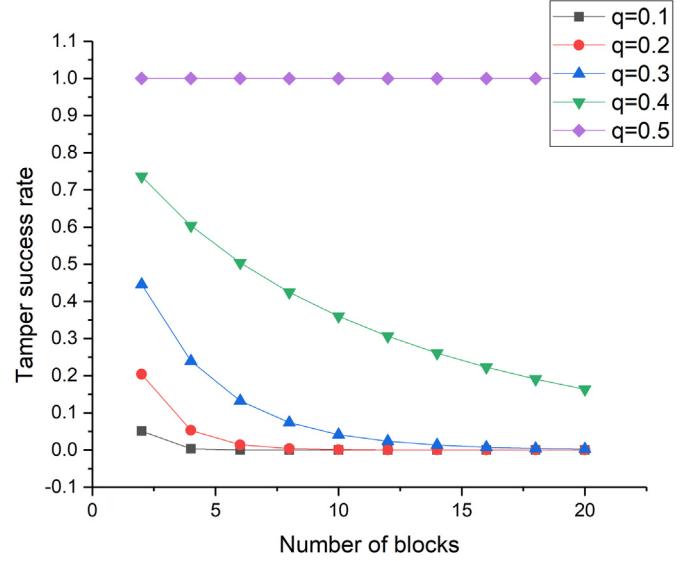


Fig. 6. Blockchain transaction risk probability assessment (q , attacker successful forgery probability).

of the blockchain system, which means launching a “51%” computing power attack. However, it is almost impossible to tamper with the data information on the consortium chain at the current computational level. The output space of SHA-512 algorithm used in the blockchain structure construction in the data acquisition layer is 2^{512} , which has a very strong anti-collision ability. Even with the supercomputer's power of “Shenwei-taihulight”, it still takes 2^{32} years to solve the SHA-512 algorithm [25]. Therefore, when the AMI data size is large enough, the block consortium chain can ensure that the AMI data will not be attacked and tampered with due to cracking technology or computing power.

4.3. Security in transaction access phase

The consortium chain has better access control function than the public chain, so the addition of new nodes will be limited, and illegal access to data will become difficult. In addition, the access operation process of data on the chain will be completely recorded in the form of a similar “log”, and the “log” cannot be modified, thus providing anti-repudiation for the system. The essence of blockchain is a distributed database. Through the application of consensus algorithm, even if some nodes in the network have problems, it will not affect the normal access to data, and the availability of data on the chain is guaranteed.

In summary, the proposed framework can meet the basic security requirements of AMI data, as shown in Table 3.

5. Performance analysis

This section first explains the configuration of platform facilities used for simulation experiments. Secondly, the performance of the proposed scheme is demonstrated by a specific example test.

Table 3
Security features comparison.

AMI data security requirements	confidential- ity	integrity	accountabil- ity	availability
Blockchain-based AMI framework for data security and privacy protection	✓	✓	✓	✓

Table 4
Performance of parameters of 1000 AMI data transactions.

Work load file	number of transactions	Maximum delay(s)	Average delay(s)	Throughput (tps)
Open	1000	2.60	1.33	62.5
Query	1000	0.13	0.02	100.1
Transfer	1000	3.82	1.95	58.2

5.1. Configuration of software and hardware parameters of tester

This paper measures the performance of deploying block networks by Hyperledger Caliper v0.3, including transaction throughput, transaction delay. The host hardware parameters used are: Intel(R) Core (TM) i7-4710MQ CPU @ 2.50 GHz 2.50 GHz. The Ubuntu16.04 system is built on the host by VMware software, and assigned to the system4 GB RAM, four processor cores.

We build a blockchain network based on Hyperledger fabric v1.4. The test network includes two organizations, each maintaining two nodes. The AMI measurement data used in the study come from the Irish social science data file [26]. The measurement data includes three parts: user ID coding, measurement time coding and power consumption. In the pressure test of the system, we simulate the reading and writing operation of the measured data by “open”, “query” and “transfer” these three workloads, and evaluate the performance of the blockchain system described in this paper by controlling the number of sending transactions.

5.2. Performance testing

Each operation request for measurement data (such as query, etc.) will generate a corresponding transaction in the blockchain network. Therefore, the performance of blockchain is mainly related to transactions.

Firstly, this paper sets the number of transactions to 1000, and the test results are shown in Table 4. It is easy to draw a conclusion that since the open and transfer operations involve changes in the world state and require the participation of consensus sorting nodes, the delay is significantly greater than the query operation, and the corresponding throughput is lower. The test results meet the expected estimates.

Next, take “open” as an example to test the impact of the change in the number of transactions on the system delay and throughput.

In the delay test, we increase the number of transactions and analyze the change of system delay. The test results are shown in Fig. 7. With the increase of the number of transactions, the system delay will also increase. After analysis, this is mainly caused by the consensus sorting algorithm. The ability of the system to process transactions in unit time is limited. When the processing speed of the system reaches the maximum, it is too late to process the subsequent transaction requests, resulting in the increase of delay. According to the description of AMI information flow characteristics in [27], the transmission mode of measurement data is unicast, and its requirement for transmission time is less than 15 s. The transmission mode of remote load control command is unicast or multicast, and the required transmission time is less than 5 s. The test results show that the average delay generated by the proposed system is about 1.33 s, which can meet the delay requirements of AMI data transmission. In the

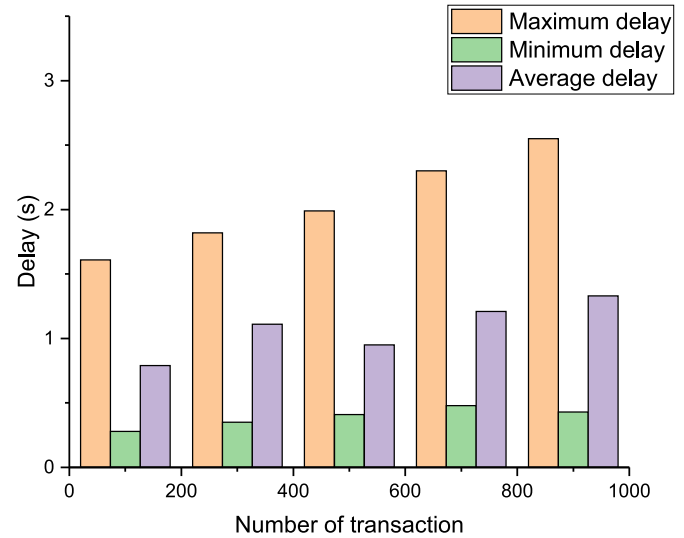


Fig. 7. Changes in delay. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

throughput test experiment, the number of transactions is set as above. The test results are shown in Fig. 8. As the number of transactions increases, the throughput of the system increases gradually. However, limited to the actual development environment, the maximum throughput of our system is only about 75 tps, while the theoretical throughput of the Hyperledger Fabric can reach about 3000 tps, which can fully respond to AMI data operation requests.

6. Conclusion

This paper provides a safe and general five-layer theoretical framework, and discusses the application of blockchain technology in AMI from a new perspective. We discuss the function and usage of each layer in the framework in detail, and verify its security and feasibility through theoretical analysis and simulation experiments.

Although the introduction of blockchain technology has a positive impact on the protection of AMI data security and privacy, because the nodes in the blockchain network are distributed, they need to use consensus algorithm to ensure the consistency of block content, which inevitably brings a lot of network communication overhead, resulting in the problem of limited system performance.

In conclusion, the proposed blockchain-based AMI data security and privacy protection framework guarantees the privacy, integrity, auditability and availability of AMI data. The experimental simulation results show that although there is an upper

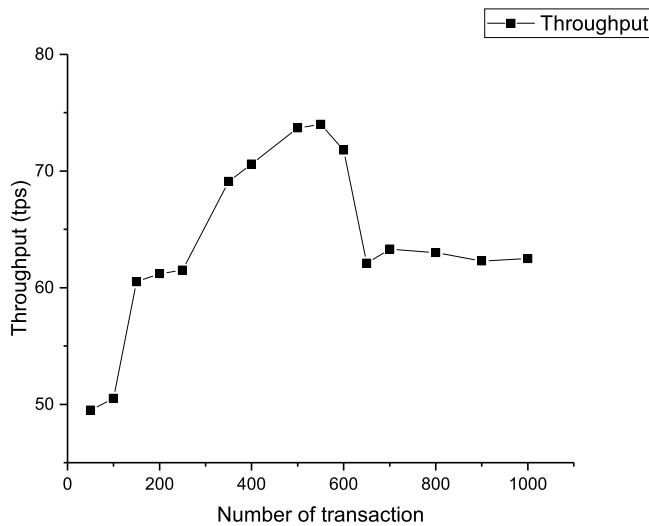


Fig. 8. Changes in throughput.

limit on the performance of the system, the current network and hardware infrastructure can meet the requirements of AMI data transaction for delay and throughput.

CRedit authorship contribution statement

Hongliang Tian: Software, Writing - review & editing. **Yuzhi Jian:** Conception to verification, to article writing, typesetting and submission. **Xiaonan Ge:** Supervision, Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

The authors thank the anonymous reviewers for their critical and constructive comments, their thoughtful suggestions have helped improve this paper substantially.

References

- [1] Z.A.E. Houda, A. Hafid, L. Khoukhi, Blockchain meets AMI: Towards secure advanced metering infrastructures, in: ICC 2020-2020 IEEE Int. Conf. Commun., ICC, IEEE, Dublin, Ireland, 2020, pp. 1–6, <http://dx.doi.org/10.1109/ICC40277.2020.9148963>.
- [2] C. Greer, D.A. Wollman, D.E. Prochaska, P.A. Boynton, J.A. Mazer, C.T. Nguyen, G.J. FitzPatrick, T.L. Nelson, G.H. Koepke, A.R. Hefner Jr., V.Y. Pillitteri, T.L. Brewer, N.T. Golmie, D.H. Su, A.C. Eustis, D.G. Holmberg, S.T. Bushby, NIST Framework and Roadmap for Smart Grid Interoperability Standards, Release 3.0, National Institute of Standards and Technology, 2014, <http://dx.doi.org/10.6028/NIST.SP.1108r3>.
- [3] S.N. Lighari, B.B. Jensen, D.M.A. Hussain, A.A. Shaikh, Attacks and their defenses for advanced metering infrastructure, in: 2014 6th Int. Congr. Ultra Mod. Telecommun. Control Syst. Workshop, ICUMT, IEEE, St. Petersburg, Russia, 2014, pp. 148–151, <http://dx.doi.org/10.1109/ICUMT.2014.7002094>.
- [4] J.C. Olivares-Rojas, J.A. Gutierrez-Gnechchi, J. Cerda-Jacobo, A transactive energy model for smart metering systems using blockchain, CSEE J. Power Energy Syst. 7 (2021) 11.
- [5] E. Zanghi, M.B. Do Coutto Filho, J.C. Stacchini de Souza, Conceptual framework for blockchain-based metering systems, Multiagent Grid Syst. 15 (2019) 77–97, <http://dx.doi.org/10.3233/MGS-190303>.
- [6] S.-K. Kim, J.-H. Huh, A study on the improvement of smart grid security performance and blockchain smart grid perspective, Energies 11 (2018) 1973, <http://dx.doi.org/10.3390/en11081973>.
- [7] H. Naseer, M.N. Mumtaz Bhutta, M.A. Alojail, A key transport protocol for advance metering infrastructure (AMI) based on public key cryptography, in: 2020 Int. Conf. Cyber Warf. Secur. ICCWS, IEEE, Islamabad, Pakistan, 2020, pp. 1–5, <http://dx.doi.org/10.1109/ICCWS48432.2020.9292385>.
- [8] Keping Yu, M. Arifuzzaman, Zheng Wen, Di Zhang, T. Sato, A key management scheme for Secure Communications of information centric advanced metering infrastructure in smart grid, IEEE Trans. Instrum. Meas. 64 (2015) 2072–2085, <http://dx.doi.org/10.1109/TIM.2015.2444238>.
- [9] J. Wang, L. Wu, K.-K.R. Choo, D. He, Blockchain-based anonymous authentication with key management for smart grid edge computing infrastructure, IEEE Trans. Ind. Inform. 16 (2020) 9.
- [10] F.M. Cleveland, Cyber security issues for advanced metering infrastructure (AMI), in: 2008 IEEE Power Energy Soc. Gen. Meet. - Convers. Deliv. Electr. Energy 21st Century, IEEE, Pittsburgh, PA, USA, 2008, pp. 1–5, <http://dx.doi.org/10.1109/PES.2008.4596535>.
- [11] C. Fan, S. Ghaemi, H. Khazaei, P. Musilek, Performance evaluation of blockchain systems: A systematic survey, IEEE Access 8 (2020) 126927–126950, <http://dx.doi.org/10.1109/ACCESS.2020.3006078>.
- [12] S.M. Farooq, S.M.S. Hussain, T.S. Ustun, A. Iqbal, Using ID-based authentication and key agreement mechanism for securing communication in advanced metering infrastructure, IEEE Access 8 (2020) 210503–210512, <http://dx.doi.org/10.1109/ACCESS.2020.3038813>.
- [13] S. Ruj, A. Nayak, A decentralized security framework for data aggregation and access control in smart grids, IEEE Trans. Smart Grid. 4 (2013) 196–205, <http://dx.doi.org/10.1109/TSG.2012.2224389>.
- [14] W. Tong, J. Liang, Z. Li, X. Jin, Security data aggregation based on data recoverability for wireless sensor network of AMI, Electric Power Autom. Equip. 37 (05) 211–216.
- [15] A. Mohammadali, M. Sayad Haghighi, M.H. Tadayon, A. Mohammadi-Nodooshan, A novel identity-based key establishment method for advanced metering infrastructure in smart grid, IEEE Trans. Smart Grid. 9 (2018) 2834–2842, <http://dx.doi.org/10.1109/TSG.2016.2620939>.
- [16] A. Kumar, B. Bhushan, P. Nand, Preventing and detecting intrusion of cyberattacks in smart grid by integrating blockchain, in: D.K. Sharma, S.-L. Peng, R. Sharma, D.A. Zaitsev (Eds.), Micro-Electron. Telecommun. Eng., Springer Singapore, Singapore, 2022, pp. 119–130, http://dx.doi.org/10.1007/978-981-16-8721-1_12.
- [17] H. Zhang, J. Wang, Y. Ding, Blockchain-based decentralized and secure keyless signature scheme for smart grid, Energy 180 (2019) 955–967, <http://dx.doi.org/10.1016/j.energy.2019.05.127>.
- [18] W. Lu, Z. Ren, J. Xu, S. Chen, Edge blockchain assisted lightweight privacy-preserving data aggregation for smart grid, IEEE Trans. Netw. Serv. Manag. 18 (2021) 1246–1259, <http://dx.doi.org/10.1109/TNSM.2020.3048822>.
- [19] R. Rashed Mohassel, A. Fung, F. Mohammadi, K. Raahemifar, A survey on advanced metering infrastructure, Int. J. Electr. Power Energy Syst. 63 (2014) 473–484, <http://dx.doi.org/10.1016/j.ijepes.2014.06.025>.
- [20] S.T. Bokhari, T. Aftab, I. Nadir, T. Bakhshi, Exploring blockchain-secured data validation in smart meter readings, in: 2019 22nd Int. Multitopic Conf. INMIC, IEEE, Islamabad, Pakistan, 2019, pp. 1–7, <http://dx.doi.org/10.1109/INMIC48123.2019.9022772>.
- [21] Y. Li, J. Meng, H. Yu, S. Wang, J. Fan, Y. Guo, Overview of Blockchain Capacity Expansion Technology, Electric Power Inf. Commun. Technol. 18 (6) 1–9.
- [22] M. Castro, B. Liskov, Practical Byzantine fault tolerance, in: Proc. Third Symp. Oper. Syst. Des. Implement, USENIX Association, USA, 1999, pp. 173–186.
- [23] D. Kraft, Difficulty control for blockchain-based consensus systems, Peer-Peer Netw. Appl. 9 (2016) 397–413, <http://dx.doi.org/10.1007/s12083-015-0347-x>.
- [24] B. Li, W. Cao, B. Qi, Y. Sun, N. Guo, Y. Su, G. Cui, Overview of Application of Block Chain Technology in Ancillary Service Market, Power Syst. Technol. 41 (3) 736–744.
- [25] J. Cui, S. Wang, Y. Xin, Research on technical framework of smart grid data management from consortium blockchain perspective, Proc. CSEE 40 (3) 836–848.
- [26] Commission for Energy Regulation (CER), Cer smart metering project - electricitycustomer behaviour trial, 2009–2010, 1st Ed., SN: 0012-00, 2012. <https://www.ucd.ie/issda/data/commissionforenergyregulationcer/>.
- [27] J. Chen, Research on Communication Model and Key Management Scheme of Smart Meter, North China Electric Power University, Hebei, 2012.